

VTPEH 6270 - Check Point 06

Statistical Analyses

Philip Aquila Salvatore Tapan Dahal

2026-04-13

Contents

1	Research Question and Background	1
2	Material and Methods	2
3	Results	4
4	Conclusions	9
5	AI Use Disclosure Statement	9
6	References	9

1 Research Question and Background

1.1 Research Question

How does non-critical food safety violations vary by county in New York State?

1.2 Scientific Plausibility and Context

County where the inspector from health department come more often to inspect the restaurants there will have more violations.¹ Restaurant inspector from health department will visit more often to counties that are more populated and have more restaurants¹ Other research showed that restaurant violations also happened more in regions with low economic status.² Low economic status usually also have low education level which can cause the restaurant workers to have less knowledge in proper way to handle food.² Other research also stated that the location of the restaurants can also relate to the violations, especially if it is located not within walking distance of a healthy source of food such as grocery stores.³

2 Material and Methods

2.1 Data

The data were collected by New York State Department of Health. They got the data by performing restaurant inspection all over New York State. This data was updated every month since January 29, 2013 and it was updated the last in February 1, 2026.

```
library(dplyr)
library(janitor)
library(knitr)
library(ggplot2)
library(crayon)
Data_Violations <- read.csv("C:/Users/Lenovo/OneDrive/Documents/Serba-Serbi Kuliah di Ithaca/Pekuliahan/
head(Data_Violations)
str(Data_Violations)
names(Data_Violations)
Data_Violations_Cleaned <- Data_Violations %>%
  dplyr::mutate(across(c(FACILITY, ADDRESS,
                        FACILITY.ADDRESS, CITY,
                        OPERATION.NAME,
                        PERMITTED...D.B.A.,
                        PERMITTED..CORP..NAME,
                        PERM..OPERATOR.LAST.NAME,
                        PERM..OPERATOR.FIRST.NAME), toupper))
Data_Violations_County <- Data_Violations_Cleaned %>%
  select(-ADDRESS, -LAST.INSPECTED, -VIOLATIONS, -DESCRIPTION,
        -X.LOCAL.HEALTH.DEPARTMENT, -CITY, -FACILITY.ADDRESS, -ZIP.CODE,
        -NYSDOH.GAZETTEER..1980., -MUNICIPALITY, -OPERATION.NAME,
        -PERMIT.EXPIRATION.DATE, -PERMITTED...D.B.A., -PERMITTED..CORP..NAME,
        -NYS.HEALTH.OPERATION.ID, -INSPECTION.TYPE, -INSPECTION.COMMENTS,
        -FOOD.SERVICE.FACILITY.STATE, -Location1, -PERM..OPERATOR.LAST.NAME,
        -PERM..OPERATOR.FIRST.NAME)
#Because I only want to compare the association between number of restaurant
#non critical violations to the counties in New York State, I just keep 5
#variables which are Facility, Total Critical Violations, Total Critical
#Violations Not Corrected, Total Noncritical Violations, and County
Data_County_Renamed <- Data_Violations_County %>%
  rename(Critical = TOTAL...CRITICAL.VIOLATIONS, Critical_Not_Corrected =
        TOTAL..CRIT...NOT.CORRECTED., Non_Critical =
        TOTAL...NONCRITICAL.VIOLATIONS)
head(Data_County_Renamed)
str(Data_County_Renamed)
names(Data_County_Renamed)
#I renamed every numerical variables to make it shorter
#Total Critical Violations becomes Critical, Total Critical Not Corrected
#becomes Critical_Not_Corrected, and Total Noncritical Violations becomes
#Non_Critical
```

2.2 Statistical Analyses

```
#install.packages("gtsummary")
library(gtsummary)
library(gt)
library(dplyr)

Data_County_Renamed %>%
  group_by(COUNTY) %>%
  summarise(
    N = n(),
    Mean = round(mean(Non_Critical, na.rm = TRUE), 2),
    SD = round(sd(Non_Critical, na.rm = TRUE), 2),
    Median = round(median(Non_Critical, na.rm = TRUE), 2),
    Q1 = round(quantile(Non_Critical, 0.25, na.rm = TRUE), 2),
    Q3 = round(quantile(Non_Critical, 0.75, na.rm = TRUE), 2),
    Missing = sum(is.na(Non_Critical))
  ) %>%
  gt() %>%
  tab_header(
    title = "Non Critical Violations by County"
  ) %>%
  cols_label(
    COUNTY = "County",
    N = "N",
    Mean = "Mean",
    SD = "SD",
    Median = "Median",
    Q1 = "Q1",
    Q3 = "Q3",
    Missing = "Missing"
  ) %>%
  gt::gtsave("my_table_revised.png")

kruskal.test(Non_Critical ~ COUNTY, data = Data_County_Renamed)

##
## Kruskal-Wallis rank sum test
##
## data: Non_Critical by COUNTY
## Kruskal-Wallis chi-squared = 3529.7, df = 54, p-value < 2.2e-16

#install.packages("dunn.test")
library(dunn.test)
dunn_result <- dunn.test(Data_County_Renamed$Non_Critical,
  g = Data_County_Renamed$COUNTY,
  method = "bonferroni",
  altp = TRUE)

Statistically_Significant_Restaurants <-
  data.frame(Comparison = dunn_result$comparisons,
    P_adjusted = dunn_result$altP.adjusted) %>%
```

```
filter(P_adjusted < 0.05)

nrow(Statistically_Significant_Restaurants)
```

```
## [1] 753
```

The mean is ranging from 0.41 - 4.18 and the median as well as the Q3 are not 0 in a lot of the counties. These tell us that the non critical violations are common and spread across restaurants and counties all over New York State. County with the highest mean and median are Cortland with 4.18 (3.05) mean and median 4 (2, 6). The mean and SD in Cortland suggest that non critical violations were more common across this county. Counties like Westchester and St. Lawrence that have similarly high mean and SD (3.62 (3.62) and 3.71 (3.6)) tell us about high variability where there are restaurants with few non critical violations and there are some with less. Those restaurants with few non critical violations pull the mean up. Counties with low mean and SD like Herkimer and Tompkins suggest that both of these counties have low violations in all of their restaurants. The variation of means tells us that a strong association between the county and non-critical violations. To tell us the significance in the association, a Kruskal-Wallis test must be done. Kruskal-Wallis test was chosen because the data is not normally-distributed and the number of restaurants are different between each counties.

The Kruskal-Wallis test shows a p value that is far below 0.05, so the H null will be rejected. This result tell us that the non critical violations have association with the counties in New York State. To know how many counties that have significant different between each other, post-hoc Dunn test will be done.

The post-hoc Dunn and bonferonni test show that there are 753 pairs of counties that have statistical difference.

2.3 Code

<https://github.com/pad238-art/R-Homework>

3 Results

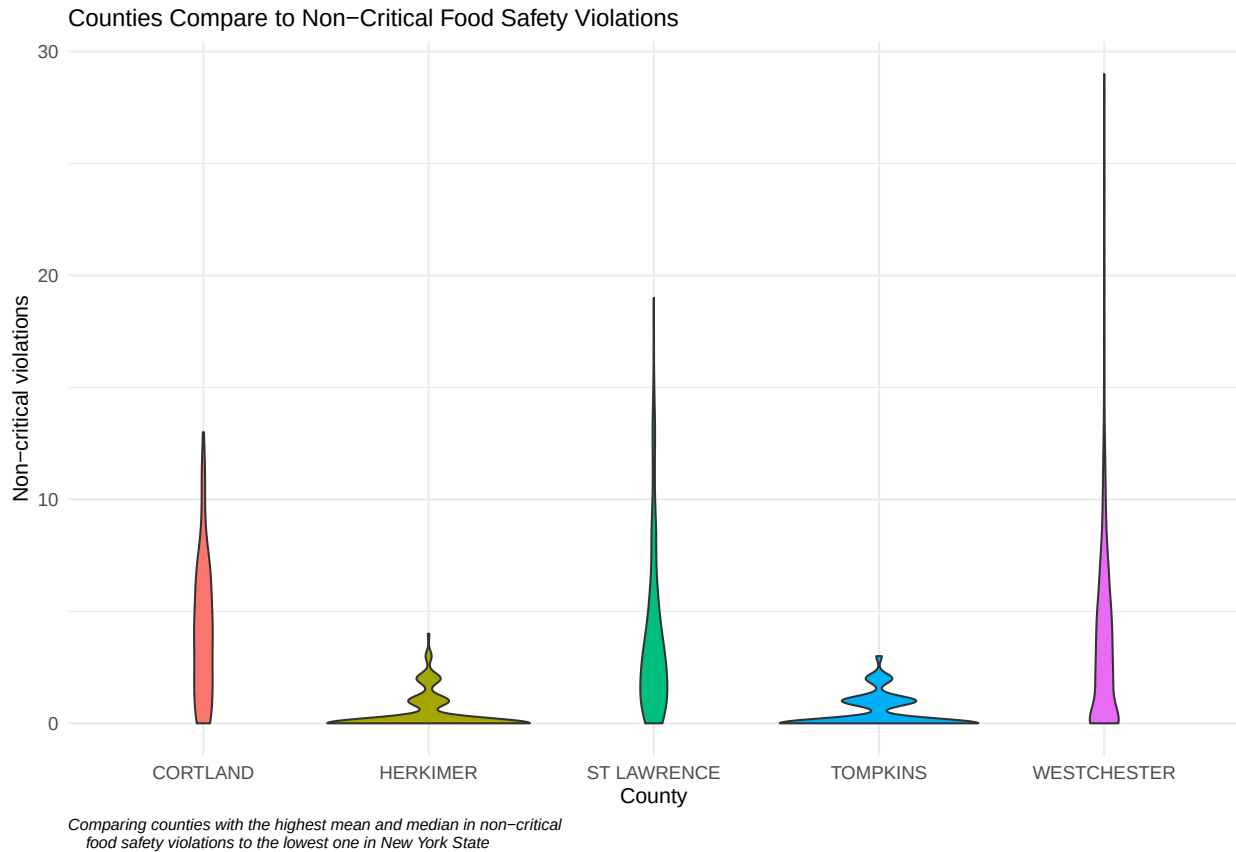
3.1 Visualization

```
Data_County_Renamed %>%
  filter(COUNTY %in% c("CORTLAND", "WESTCHESTER", "ST LAWRENCE",
                      "HERKIMER", "TOMPKINS")) %>%
  ggplot(aes(x = COUNTY, y = Non_Critical, fill = COUNTY)) +
  geom_violin() +
  labs(
    title = "Counties Compare to Non-Critical Food Safety Violations",
    x = "County",
    y = "Non-critical violations",
    caption = "Comparing counties with the highest mean and median in non-critical
              food safety violations to the lowest one in New York State"
  ) +
  theme_minimal(base_size = 8) +
  theme(
    axis.text = element_text(size = 7),
    axis.title = element_text(size = 8),
```

```

plot.title = element_text(size = 9),
plot.caption = element_text(size = 6, hjust = 0, face = "italic"),
legend.position = "none"
)

```



#Violin plot comparing counties with the highest mean and median to the lowest one

```

range(as.Date(Data_Violations$LAST_INSPECTED, format = "%m/%d/%Y"), na.rm = TRUE)

```

```

## [1] "2007-03-23" "2026-01-30"

```

#The time frame is 2007-2026

```

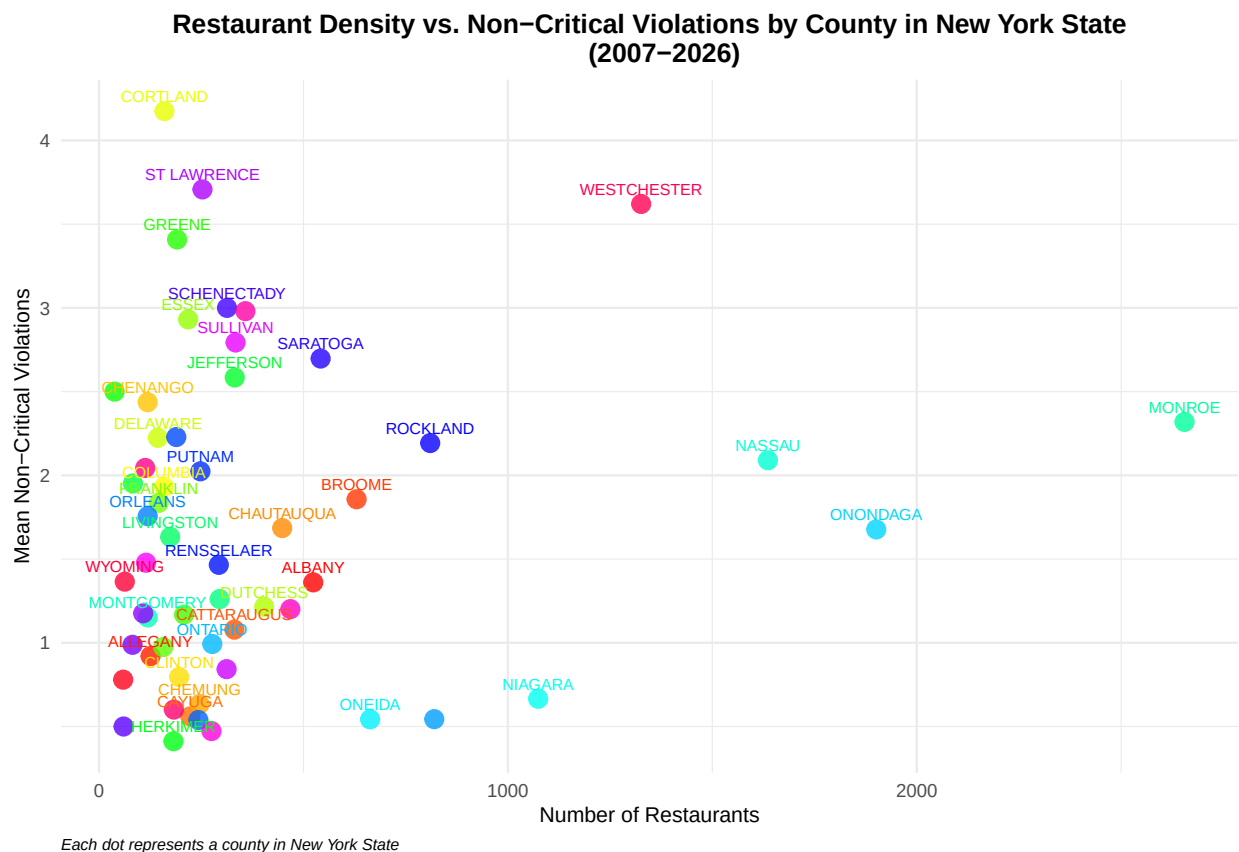
Data_County_Renamed %>%
  group_by(COUNTY) %>%
  summarise(
    Mean_Non_Critical = mean(Non_Critical, na.rm = TRUE),
    Num_Restaurants = n()
  ) %>%
  ggplot(aes(x = Num_Restaurants, y = Mean_Non_Critical, color = COUNTY)) +
  geom_point(size = 3, alpha = 0.8) +
  geom_text(aes(label = COUNTY), size = 2, vjust = -0.8, check_overlap = TRUE) +
  scale_color_manual(values = rainbow(55)) +
  labs(

```

```

title = "Restaurant Density vs. Non-Critical Violations by County in New York State
(2007-2026)",
x = "Number of Restaurants",
y = "Mean Non-Critical Violations",
caption = "Each dot represents a county in New York State"
) +
theme_minimal(base_size = 9) +
theme(
  plot.title = element_text(size = 10, face = "bold", hjust = 0.5),
  axis.text = element_text(size = 7),
  axis.title = element_text(size = 8),
  plot.caption = element_text(size = 6, hjust = 0, face = "italic"),
  legend.position = "none"
)

```



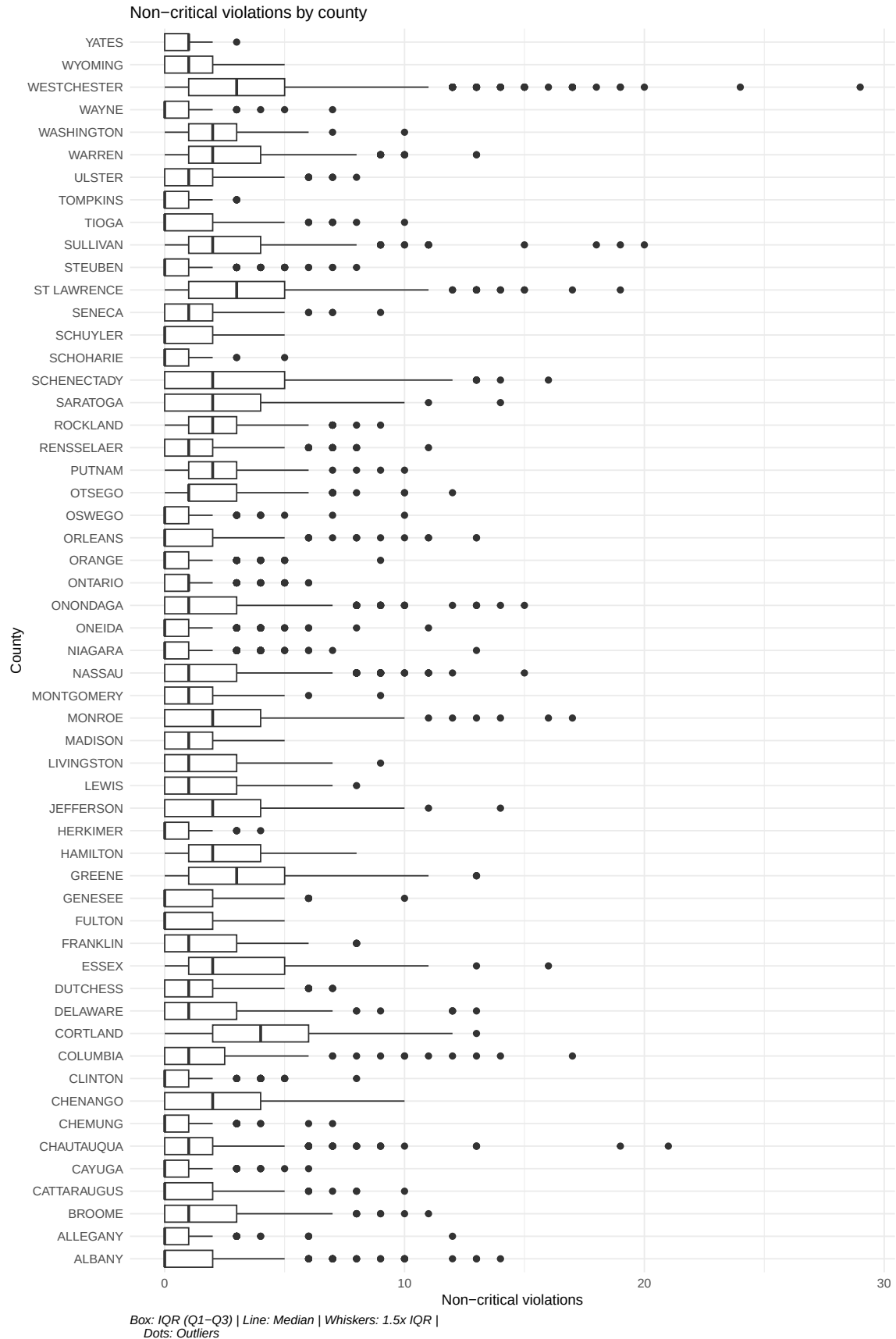
#The trend is more restaurants, more non-critical food safety violations

```

ggplot(Data_County_Renamed, aes(x = COUNTY, y = Non_Critical)) +
  geom_boxplot() +
  coord_flip() +
  labs(
    title = "Non-critical violations by county",
    x = "County",
    y = "Non-critical violations",
    caption = "Box: IQR (Q1-Q3) | Line: Median | Whiskers: 1.5x IQR |

```

```
  Dots: Outliers"
) +
theme_minimal(base_size = 7) +
theme(
  axis.text.y = element_text(size = 6, lineheight = 1.5),
  axis.text.x = element_text(size = 6),
  axis.title = element_text(size = 7),
  plot.title = element_text(size = 8),
  plot.caption = element_text(size = 6, hjust = 0, face = "italic")
)
```



3.2 Statistical Analyses

In almost every counties, non-critical food safety violations following right-skewed distribution-most restaurants have zero or very little amount of violations, but few restaurants that have a lot of violations were pulling the mean higher which happened in most of the counties. This tells us that more about systemic characteristic of food safety violations instead of county-specific phenomenon.

4 Conclusions

The result on the scatter plot shows us that the trend is more restaurants in the county, more non critical restaurant violations that the county would have. This is align with the result that Harris, et al. found where they mentioned that restaurant inspector from the health department will come more to inspect in places that have a lot of people and restaurants in it, hence the higher food safety violations they will find.¹ Further research can also be done to analyze if it has relationship with economic and education status like what Sha, et al. found in their research where more restaurant violations found in places populated with people that have lower economic and education.²

5 AI Use Disclosure Statement

I did not use AI for this Checkpoint

6 References

1. Harris KJ, DiPietro RB, Murphy KS, Rivera G. Critical Food Safety Violations in Florida: Relationship to Location and Chain vs. Non-Chain Restaurants. *Int J Hosp Manag.* 2014;38:57-64. doi:10.1016/j.ijhm.2013.12.005
2. Sha Y, Song X, Zhan J, Lu L, Zhang Q, Lu Y. Regional Character, Restaurant Size, and Food Safety Risk: Evidence from Food Safety Violation Data in Gansu Province, China. *J Food Prot.* 2020;83(4):677-685. doi:10.4315/0362-028X.JFP-19-457
3. Parsa HG, Kreeger JC, van der Rest JP, Xie L “Karen”, Lamb J. Why Restaurants Fail? Part V: Role of Economic Factors, Risk, Density, Location, Cuisine, Health Code Violations and GIS Factors. *Int J Hosp Tour Adm.* 2021;22(2):142-167. doi:10.1080/15256480.2019.1598908